

## MAEBASHI, Japan

WMO: 476240

Lat: 36.4053N

Lon: 139.0606E

Elev: 114

StdP: 99.97

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -2.6       | -1.6 | -13.1                         | 1.2 | 5.1  | -11.7 | 1.4 | 4.1  | 9.1                   | 6.7  | 8.2 | 6.2  | 2.9                   | 320  | 0.396 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.1                    | 35.2            | 25.0 | 33.7 | 24.6 | 32.2 | 24.1 | 26.0                | 32.4 | 25.4 | 31.5 | 24.9 | 30.5 | 3.4                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.3                            | 19.6 | 28.8 | 23.7 | 18.8 | 28.4 | 23.2 | 18.2 | 28.0 | 81.3          | 32.3 | 78.8 | 31.7 | 76.4 | 30.7 | 28.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 7.7           | 6.6  | 5.8 | -4.6                       | 37.6 | 1.1                | 1.0 | -5.4                                               | 38.3 | -6.0       | 38.9 | -6.6       | 39.4 | -7.4       | 40.2 |    |    |
| (5)               |      |     | -6.4                       | 26.9 | 0.9                | 0.6 | -7.0                                               | 27.3 | -7.5       | 27.7 | -8.0       | 28.1 | -8.6       | 28.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 15.4   | 4.1   | 4.8   | 8.1   | 13.7  | 18.9  | 22.3  | 26.3  | 27.3  | 23.2  | 17.5  | 11.5  | 6.6  |
|                                                                                  | DBStd       | 8.58   | 2.24  | 2.66  | 3.23  | 3.54  | 2.97  | 2.82  | 3.13  | 2.78  | 3.21  | 2.94  | 2.91  | 2.67 |
|                                                                                  | HDD10.0     | 550    | 184   | 149   | 79    | 9     | 0     | 0     | 0     | 0     | 0     | 0     | 17    | 112  |
|                                                                                  | HDD18.3     | 1938   | 443   | 380   | 318   | 143   | 29    | 3     | 0     | 0     | 2     | 51    | 205   | 365  |
|                                                                                  | CDD10.0     | 2527   | 1     | 2     | 19    | 120   | 276   | 368   | 507   | 537   | 397   | 233   | 62    | 5    |
|                                                                                  | CDD18.3     | 873    | 0     | 0     | 0     | 4     | 46    | 122   | 248   | 279   | 149   | 26    | 0     | 0    |
|                                                                                  | CDH23.3     | 7676   | 0     | 0     | 1     | 41    | 332   | 817   | 2436  | 2912  | 1044  | 93    | 0     | 0    |
|                                                                                  | CDH26.7     | 2983   | 0     | 0     | 0     | 3     | 68    | 236   | 1063  | 1278  | 323   | 12    | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.8    | 3.2   | 3.3   | 3.4   | 3.1   | 2.7   | 2.3   | 2.2   | 2.3   | 2.3   | 2.5   | 2.8   | 3.1  |
| Precipitation                                                                    | PrecAvg     | 1182   | 25    | 31    | 53    | 75    | 97    | 156   | 183   | 187   | 194   | 112   | 45    | 22   |
|                                                                                  | PrecMax     | 1659   | 105   | 158   | 122   | 174   | 255   | 328   | 376   | 558   | 450   | 393   | 198   | 115  |
|                                                                                  | PrecMin     | 796    | 0     | 0     | 1     | 5     | 21    | 45    | 45    | 24    | 21    | 6     | 0     | 0    |
|                                                                                  | PrecStd     | 200    | 25    | 30    | 29    | 37    | 49    | 67    | 73    | 96    | 100   | 79    | 36    | 23   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 14.4  | 17.3  | 21.9  | 26.8  | 30.8  | 33.6  | 36.8  | 36.9  | 33.7  | 28.4  | 21.8  | 17.1 |
|                                                                                  |             | MCWB   | 7.2   | 9.6   | 11.1  | 14.6  | 18.3  | 22.3  | 25.1  | 25.4  | 24.0  | 19.9  | 14.1  | 10.4 |
|                                                                                  | 2%          | DB     | 11.7  | 13.6  | 18.5  | 24.4  | 28.5  | 30.9  | 35.3  | 35.4  | 31.7  | 25.7  | 19.8  | 14.6 |
|                                                                                  |             | MCWB   | 5.4   | 6.7   | 9.9   | 14.0  | 17.4  | 20.9  | 25.0  | 25.2  | 23.3  | 18.5  | 13.1  | 8.5  |
|                                                                                  | 5%          | DB     | 10.0  | 11.6  | 16.2  | 22.3  | 26.8  | 29.3  | 33.8  | 34.0  | 30.1  | 23.9  | 18.1  | 12.9 |
|                                                                                  |             | MCWB   | 4.2   | 5.4   | 8.5   | 13.3  | 16.6  | 20.4  | 24.7  | 24.9  | 22.7  | 17.3  | 12.0  | 7.4  |
|                                                                                  | 10%         | DB     | 8.6   | 9.9   | 14.1  | 20.3  | 25.1  | 27.7  | 32.1  | 32.6  | 28.5  | 22.4  | 16.6  | 11.3 |
|                                                                                  |             | MCWB   | 3.3   | 4.3   | 7.9   | 12.5  | 16.2  | 19.9  | 24.2  | 24.6  | 21.9  | 16.5  | 11.2  | 6.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 8.2   | 10.4  | 12.9  | 18.6  | 21.3  | 24.3  | 26.7  | 27.0  | 25.3  | 21.6  | 16.1  | 11.7 |
|                                                                                  |             | MCDB   | 12.0  | 16.3  | 17.8  | 22.1  | 26.3  | 30.3  | 33.8  | 34.0  | 30.6  | 25.9  | 19.1  | 14.9 |
|                                                                                  | 2%          | WB     | 6.2   | 7.7   | 11.3  | 16.3  | 19.6  | 23.1  | 25.9  | 26.2  | 24.5  | 20.1  | 14.6  | 9.6  |
|                                                                                  |             | MCDB   | 10.9  | 12.1  | 16.5  | 22.0  | 25.2  | 28.2  | 32.6  | 32.5  | 29.3  | 24.2  | 17.6  | 13.2 |
|                                                                                  | 5%          | WB     | 4.7   | 6.1   | 9.8   | 14.8  | 18.4  | 22.1  | 25.3  | 25.7  | 23.8  | 18.8  | 13.4  | 8.1  |
|                                                                                  |             | MCDB   | 9.2   | 10.4  | 14.9  | 19.5  | 23.6  | 27.0  | 31.6  | 31.7  | 28.1  | 22.3  | 16.5  | 12.2 |
|                                                                                  | 10%         | WB     | 3.6   | 4.7   | 8.3   | 13.5  | 17.4  | 21.2  | 24.7  | 25.2  | 23.1  | 17.7  | 12.2  | 6.7  |
|                                                                                  |             | MCDB   | 7.9   | 9.3   | 13.3  | 18.3  | 22.7  | 25.6  | 30.5  | 30.9  | 27.1  | 21.3  | 15.6  | 10.7 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.7   | 9.1   | 9.6   | 10.3  | 9.7   | 8.1   | 7.9   | 8.1   | 7.4   | 7.9   | 8.7   | 8.8  |
|                                                                                  |             | MCDBR  | 10.6  | 11.7  | 12.8  | 13.4  | 12.9  | 11.0  | 10.7  | 10.5  | 9.4   | 9.6   | 10.3  | 10.3 |
|                                                                                  |             | MCWBR  | 6.3   | 7.0   | 7.3   | 6.9   | 5.9   | 4.5   | 3.8   | 3.7   | 3.7   | 4.8   | 5.8   | 6.2  |
|                                                                                  | 5% WB       | MCDBR  | 9.9   | 10.6  | 11.0  | 10.3  | 9.9   | 8.5   | 9.6   | 9.3   | 8.0   | 7.7   | 8.2   | 9.4  |
|                                                                                  |             | MCWBR  | 6.3   | 7.0   | 7.1   | 6.5   | 5.6   | 3.9   | 3.6   | 3.5   | 3.5   | 4.4   | 5.4   | 6.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.373  | 0.420 | 0.494 | 0.550 | 0.583 | 0.627 | 0.620 | 0.621 | 0.534 | 0.478 | 0.410 | 0.373 |      |
|                                                                                  | taud        | 2.110  | 1.976 | 1.869 | 1.823 | 1.799 | 1.764 | 1.834 | 1.823 | 2.002 | 2.075 | 2.190 | 2.198 |      |
|                                                                                  | Ebn at Noon | 787    | 791   | 767   | 749   | 731   | 698   | 700   | 686   | 726   | 728   | 745   | 762   |      |
|                                                                                  | Edh at Noon | 119    | 154   | 188   | 208   | 216   | 224   | 207   | 206   | 163   | 139   | 111   | 103   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.50   | 3.22  | 4.08  | 4.79  | 5.21  | 4.69  | 4.66  | 4.65  | 3.75  | 3.11  | 2.51  | 2.22  |      |
|                                                                                  | RadStd      | 0.14   | 0.21  | 0.24  | 0.40  | 0.52  | 0.35  | 0.70  | 0.48  | 0.32  | 0.30  | 0.20  | 0.15  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (43 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page